SENTINEL

Sensors as processes: fork, exec, pipes, a FIFO and a scheduler

Prelab 2

Status and deadline

This prelab is **required**, and it has **no marks**. Submit it on Brightspace by **Wednesday, October 28, 2026, by midnight**. The Lab 2 session is on Friday, October 30, 2026.

The oral exam of Lab 2[1] can ask you each question of this prelab. The deliverable of Lab 2 is the code that these questions describe: the Lab 2 Brief[2] states it. The course marks the deliverable out of 80, and it gives 5% of the course grade. The course marks the oral exam out of 20, and it gives 3%. Together they give 8%.

Instructions

In Lab 2, each sensor runs as a process of its own, and it sends its reports to the centre through a pipe. The function `sensor_spawn` starts one sensor. It makes a pipe and calls `fork()`. The child calls `dup2()` to put the write end of the pipe on descriptor 1, and then `execvp()` to run `contactfeed`.[3] Each question covers one part of that work.

- Do this prelab alone, and write the answers in your own words.
- Each question names the manual pages that you need. If they are not on your machine, install them with `sudo apt install manpages manpages-dev`.
- You do not need the code of Lab 2. Each other fact that you need is in the question.
- You can run a program to check an answer. Give your answer first.

Q1. `fork()` and the buffer of `stdout`

Read `man 2 fork`, `man 3 setbuf`, `man 3 exit` and `man 2 _exit`. Look at this program:

```
int main(void)
{
    printf("SENTINEL starting\n");
    pid_t pid = fork();
    if (pid == 0) {
        return 0;               /* a return from main calls exit() */
    }
    waitpid(pid, NULL, 0);
    return 0;
```

[1] The oral exam of Lab 2 is in the Lab 3 session. The Submission Guide, Section 3, `docs/submit/submit.pdf`, describes its format.

[2] The Lab 2 Brief, `docs/lab2/lab2_brief.pdf` in your repository. It states the contract of Lab 2, its ideas and its hand-in.

[3] `contactfeed` is the given program that writes contact reports in place of the four sensors. The SENTINEL Lab Manual, Section 3.1, `docs/manual/sentinel_manual.pdf`, describes it.

```
}
```

(a) How many times does `SENTINEL starting` appear when standard output is a terminal? How many times when it is a file, with `./a.out > out.txt`? Explain the difference. What must `sensor_spawn` do before `fork()`?

(b) `execvp()` fails when the program does not exist. The child then calls `_exit(127)`, not `exit(127)`. At that time, descriptor 1 of the child is the pipe to the centre. Suppose that `sensor_spawn` has no `fflush()`. What can `exit()` do that `_exit()` does not do, and what does the centre then read?

Q2. End of file, on a pipe and on a FIFO

Read `man 7 pipe`, `man 7 fifo` and the part about `POLLHUP` in `man 2 poll`.

(a) A parent makes a pipe: `fd[0]` is its read end, and `fd[1]` is its write end. Then it calls `fork()`. The child closes `fd[0]`, writes one report to `fd[1]`, and exits. The parent does not close `fd[1]`. It reads `fd[0]` in a loop until `read()` gives 0. What does the loop do after it reads the report? Why?

(b) The centre also reads a FIFO: a pipe with a name, which any program can open to send an urgent contact. A sender opens the FIFO, writes one line and closes it. After that, no program has it open to write. Then a `read()` of the FIFO gives 0 at once, and `poll()` marks the FIFO each time. Why is that a fault for a centre that waits in `poll()`? The centre opens a write end of its own FIFO and keeps it open (descriptor 5 in the table of Q3). How does that remove the fault?

Q3. What `exec` keeps

Read `man 2 execve`, `man 2 dup2`, and the flag `O_CLOEXEC` in `man 2 open`. Before it starts the NWS sensor, the centre holds these descriptors:

fd	What it is
0, 1, 2	the terminal
3	the mission log[4]
4, 5	the FIFO: its read end, and a write end
6	the read end of the new pipe
7	the write end of the new pipe

The child calls `dup2(7, 1)`, and then `execvp()` runs `contactfeed`.[5]

(a) Each descriptor 3 to 7 has `O_CLOEXEC`. Which descriptors does `contactfeed` hold, and what is each one? Descriptor 7 has the flag: why does descriptor 1 stay open?

(b) No descriptor has `O_CLOEXEC`. Which descriptors does `contactfeed` hold now? Give one fault that this causes.

[4] The mission log is the file in which SENTINEL records each action, with its time (requirement R6). The Lab 1 Brief, Section 1, `docs/lab1/lab1_brief.pdf`, states what Lab 1 writes to it.

[5] `contactfeed` is the given program that writes contact reports in place of the four sensors. The SENTINEL Lab Manual, Section 3.1, `docs/manual/sentinel_manual.pdf`, describes it.

Q4. Collect each child

Read `man 2 waitpid` and `man 7 signal`. On Linux, `waitpid()` stores one `int`, the status. When the child exits, bits 8 to 15 hold its exit code. When a signal kills the child, bits 0 to 6 hold the number of the signal.

(a) `waitpid()` gives the status `9` for one child, and the status `32512` for another. For each one, which of `WIFEXITED` and `WIFSIGNALED` is true, what does `WEXITSTATUS` or `WTERMSIG` give, and what happened to the child?

(b) The centre has other children than its sensors. For example, it can start a helper process that sends one urgent contact, and other code collects that helper with `waitpid()` itself. The function `sensor_reap` calls `waitpid(pid, &status, WNOHANG)` once for each sensor, and not `waitpid(-1, &status, WNOHANG)`. What fault does `-1` cause?

Q5. Who goes next

Reports wait in the centre until it can process them. Its scheduler chooses the next one with a policy. The class of a report is routine, `P` (the flag of the sensor) or HIGH (the class that the centre gives).[6]

- **FCFS** (first come, first served) takes the oldest report.
- **Priority** takes a HIGH report, then a `P` report, then a routine report. In each class, it takes the oldest.
- **Round robin** takes the sensors in turn, in the order NWS, AUR, AIS, SAT, and starts with NWS. It takes the oldest report of a sensor, and skips a sensor that has no report.

(a) Eight reports wait in this order. Give the order in which the centre takes them under each policy.

Report	A1	S1	S2	N1	S3	U1	A2	S4
Sensor	AIS	SAT	SAT	NWS	SAT	AUR	AIS	SAT
Class	routine	P	routine	HIGH	P	routine	routine	routine

(b) Which policy can make the routine reports of AIS wait for ever, and when? Why can the other two not do that?

[6] HIGH is the priority that the centre gives: a contact that is on no list, in a region that the centre monitors. `P` is the flag of the sensor. The SENTINEL Lab Manual, Section 1.4, `docs/manual/sentinel_manual.pdf`, describes both.